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ACUTE PAIN

YGAD OPG 018 V 1.0 OCTOBER 2024.

Guideline on the pain assessment tools used by the Acute Pain Service

REVIEW BY: OCTOBER 2027

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KEY POINTS

- Valid and reliable pain assessment tools are vital for good clinical care to ensure accurate and reproducible results.
- The key components of assessment are the pain history, measures of pain severity, the functional impact of pain, and response to treatment.
- Neuropathic pain should be suspected if a patient whose pain appears to be responding poorly to opioids, especially if they are sedated: the continued use of opioids in these circumstances may result in patient harm.
- Safe and effective use of opioid medications requires individualisation of the opioid regime according to a patient's functional activity rather than unidimensional pain scores alone, as well as the onset of adverse effects.
- Anxiety, depression, pain catastrophizing, and emotional distress are associated with higher pain intensity.
- Regular assessment of a patient's level of sedation is more reliable clinical indicator or early OIVI than a decrease in respiratory rate.

Guideline on pain assessment tools used by the Acute Pain Service

Introduction

Valid and reliable pain assessment tools are vital for good clinical care to ensure accurate and reproducible results. The key component of assessment are the pain history, general medical history and examination, review if relevant investigations, measures of the pain severity and the functional impact of pain, and the response to treatment. There also needs to be an understanding of some of the psychological and environmental factors that may contribute to the reported experience of pain (Macintyre & Schug, 2021).

It is now recognised that unidimensional patient-reported pain scores are only one aspect of assessing a patient's pain and their response to treatment and their response to treatment, more important, an assessment of how pain is affecting the patient's physical function should also be made (Joint Commission, 2017, Levy et al, 2018; Schug et al., 2020). Doses of analgesic medication, especially opioids, should never be increased on the basis of a patient's pain scores alone.

This guidance applies to the following groups of staff who are involved in assessing and monitoring inpatients in acute pain:

ASSESSMENT OF PAIN AND PAIN RELIEF

The key components of assessment are the pain history, general medical history and examination, review of relevant investigations, measures of pain severity and the functional impact of pain, and response to treatment. There also need to be an understanding of some of the psychological and environmental factors that may contribute to the reported pain experience (Appendix 1)

MEASUREMENTS

A number of simple clinical techniques are available for the assessment and measurement of pain and its response to treatment. These are used when the patient is at rest as well as physical activity such as coughing, deep breathing, or walking. Pain and the response to treatment is assessed regularly during the treatment period. The frequency of this assessment will vary according to the analgesic technique chosen. The frequency is increased if the pain is poorly controlled or if the pain stimulus or treatment intentions are changing.

Whatever method of pain assessment is chosen, it is the trajectory of the measure used, rather than the actual number of descriptions given by the patient, that may give a better indication of patient progress. If the trajectory is not decreasing, that is, either not resolving or is increasing, a repeat pain history will help to determine if the nature of pain has changed, if there is a new cause for the pain (for example, post operative or other complication, or the presence of neuropathic pain), or whether a change should be made to the analgesic regime.

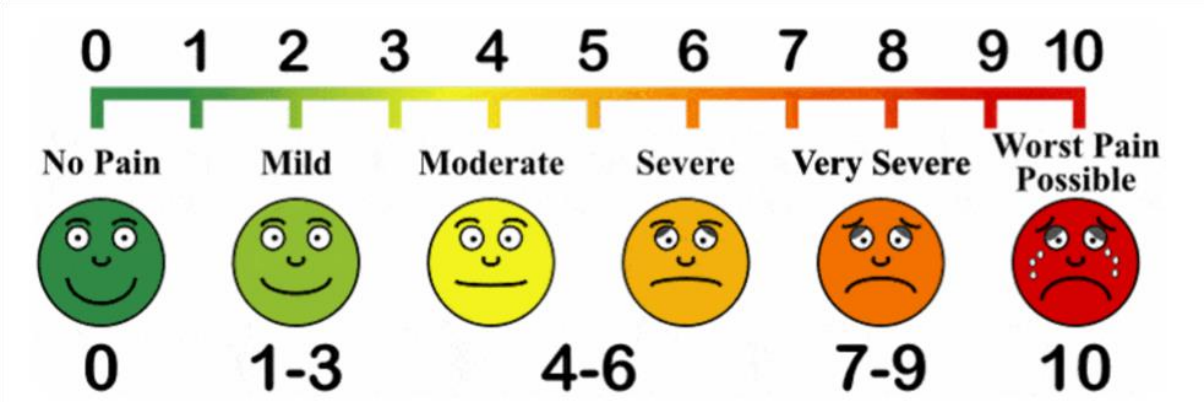
Unidimensional Measures

In adults, three common self-report measures of pain intensity (a single dimension of pain) are the visual analogue scale (VAS), the verbal numerical rating scale (VNRS) and the verbal descriptor scale (VDS). Each of these methods is reasonably reliable as long as end points and adjectives are carefully selected and standardised. While often used to compare levels of pain between patients, these methods of scoring

pain are probably of most use when measuring changes within each patient. There is good correlation between VAS, VNRS and VDS (Schug et al, 2020).

Visual Analog Scale

The VAS uses a 10-cm line with endpoint descriptors such as “no pain” marked at the left end of the line and “worst pain possible” marked at the right end. There are no other cues marked on the line. The patient is asked to mark a point on the line that best represents their pain. The distance from “no pain” to the patient’s mark is then measured in millimetres – this is the VAS score (0-100).



The disadvantage of the VAS scale is that is can be more time consuming than other simple methods, specific equipment is required (although only a pen and paper or a ruler), and some patients may find it difficult understanding and performing their score. One advantage is the wording can be written in many languages. The VAS scale can also be adapted to measure other variables such as patients’ satisfaction, side effects such as nausea and vomiting, and the degree of pain relief. The end point of a VAS for the later would be “no relief” and “complete relief”. The numerical rating scale (NRS) a calibrated VAS with the numbers 0 – 10 marked on a horizonal line is also sometimes used.

Verbal Numerical Rating Score

The VNRS, which is similar to the VAS, is the easiest to administer and correlates well with the VAS (Karcioglu et al, 2018). Patients are asked to imagine that “*0 equals no pain* “and “*10 equals the worst pain possible*”. And then to give a number that would represent their pain. Similarly, they could be asked to imagine that “0 equals no pain relief” and “10 equals complete pain relief”. The advantage of this type of system is that is does not require equipment. However, problems occur if there is a language barrier.

Verbal Descriptor Scale

The verbal descriptor scale uses different words to rate the severity of pain such as *none, mild, moderate, severe,* and *excruciating.* These scales are quick and easy to use and may be more reliable in some patients (e.g., older patients). A VDS can also be used to measure pain relief with words such as *none, slight, moderate, good* and *complete.*

What degree of pain relief is desirable?

It is not possible, practical, or safe to aim for complete pain relief at all times with most of the drugs and drug administration techniques used in the treatment of acute pain. The aim of treatment should be patient comfort, both at rest and physical activity, and good function, rather than low pain or no pain. Therefore, alterations in analgesic regimes need to take into account not just the patients' pain scores and the level they would regard as comfortable but their functional activity. The presence or absence of side effects from analgesic drugs will also affect what alterations are made to treatment orders.

Discrepancies between pain behaviour and a patient's self-report of pain may result from different coping skills. Staff should not assume, for example, that a patient who is reading or sleeping is comfortable.

High pain scores may not always require the dose of analgesic to be increased. This does not mean the patient's report of pain is to be disbelieved but that the appropriate therapeutic response to the reported pain may vary.

A reduction in pain intensity of 30%-35% has been rated clinically meaningful by patients with acute pain from a variety of sources (e.g., after surgery, trauma, and acute cancer pain) (Schug et al, 2020).

Assessment of Function

Measurements of pain intensity is only one part of the evaluation of analgesic adequacy. Assessment of function for example, the ability to take deep breaths, cough, and ambulate after surgery or injury gives an important indication of the effectiveness of pain relief. Changes in acute pain management regimes should be graded by multiple measures of the patient's pain experience, including the assessment of the patient's performance of necessary activities, rather than unidimensional pain scores alone (von Boekel, 2017). An analgesic regime should not presume to have "failed" solely on the basis of reported high pain scores, as there may be many reasons for this. For example, predictors of high postoperative pain scores include psychological comorbidities, pre-existing chronic pain and preadmission opioid intake (Yang et al., 2019, Schnabel et al, 2020). However,

functional limitation because of pain means a re-evaluation of therapy is required (Appendix 2.)

A number of assessment tools are available to measure the functional impact of pain in the chronic setting. However, a simple way to document functional ability in a patient with acute pain is the use the functional activity score (FAS) (Scott & McDonald, 2008). It is designed to assess using a "score" of A, B or C, whether a patient's current level of pain relief enables them to undertake a relevant activity (e.g., deep breathing and coughing following a laparotomy, knee flexion after knee surgery), and act as a trigger for intervention should this not be the case. The score is assessed on the basis of limitation due to the "new" acute pain and not any pre-existing baseline restrictions,

Assessment of Neuropathic Pain

The unidimensional tools described previously are not adequate when it comes to identifying or quantifying neuropathic pain. In the acute pain setting, diagnosis often relies on a clinical assessment of the patient and recognition of the signs and symptoms that are typical neuropathic pain. In this setting specifically, response to medications (e.g., poor response to opioids, good response to anti-neuropathic agents). Can be a useful indicator of the presence of neuropathic pain, as well as listening to the patient's description of pain, common descriptors include these spontaneous, shooting ("electric-shock like") or burning pain, dysesthesia ("pins and needles," "ants crawling"), allodynia, and hyperalgesia.

A variety of screening tools assessing such verbal descriptors of neuropathic pain, with or without examination of the patient, have been developed to assist with its diagnosis. Most can identify patients with neuropathic pain with relatively high specificity and sensitivity but cannot replace clinical assessment and judgement. Tools that have been validated in the diagnoses of neuropathic pain in general include Douleur Neuropathique on 4 questions (DN4), the Leeds Assessment of Neuropathic Symptoms and Sign (LANSS), the Neuropathic Pain Questionnaire (NPQ) ID-Pain, and the PainDetect questionnaire (Schug et al, 2020). Many have been validated in a range of languages. The Acute Pain Team uses the 'Leeds Assessment of Neuropathic Symptoms and Sign (LANSS)' is use (Appendix 3)

Assessment of pain in patients with communication of cognitive impairment

In some patients it may not be possible to obtain reliable self-reports of pain (e.g., when there are issues with communication due to language difficulties or cognitive impairment). In such patients, alternative measures of pain are required. Such as an assessment of patient's behaviours (e.g., grimacing, groaning, guarding and rubbing) or observing physiological responses to pain (e.g., increases in blood pressure, heart rate and sweating.

In older patients, a good correlation has been shown between unidimensional and behavioural measures in those who are cognitively intact, but it is not known if the same correlation exists in those with cognitive impairment. Unfortunately, these changes in behaviour are not unique to acute pain, and assessment of pain by observation of patient behaviour and/or vital signs should be reserved only for situations when self-report measures cannot be used.

Examples of some of the many observational pain assessments measures that have been developed to assist in the assessment of pain in patients assessment measures that have been developed to assist in the assessment of pain in patients who are unable to self-report include *Pain Assessment in Advanced Dementia* (PAINAD) (Appendix 4), *Pain Assessment Checklist for Seniors with Limited Ability to Communicate* (PASLAC), the *Abbey Pain Scale, the Faces Pain Scale, and the Mobilisation-Observation-Behaviour-Intensity-Dementia Pain Scale* (Schug et al, 2020).

Assessment of children

Tools used for pain assessment have been selected on their validity, reliability and usability and are recognized by pain specialists to be clinically effective in assessing acute pain. All share a common numeric and recorded as values 0-10 and documented on the clinical observation chart as the 5th vital sign.
The importance of using the same numeric value (0-10) is that the number relates to the same pain intensity in each tool.

Three ways of measuring pain:

- Self-report - what the child says (the gold standard)
- Behavioural –how the child behaves
- Physiological –clinical observation

1. The Wong Baker Faces Pain Scale (https://wongbakerfaces.org/)

How to use?
This tool was created for children aged 3 and older, to help them communicate about their pain.

Explain to the person that each face is for a person who feels happy because he has no pain (hurt) or sad because he has some or a lot of pain. **Face 0** is very happy because he doesn't hurt at all. **Face 2** hurts just a little bit. **Face 4** hurts a little more. **Face 6** hurts even more. **Face 8** hurts a whole lot. **Face 10** hurts as much as you can imagine, although you don't have to be crying to feel this bad. Ask the person to choose the face that best describes how he is feeling.

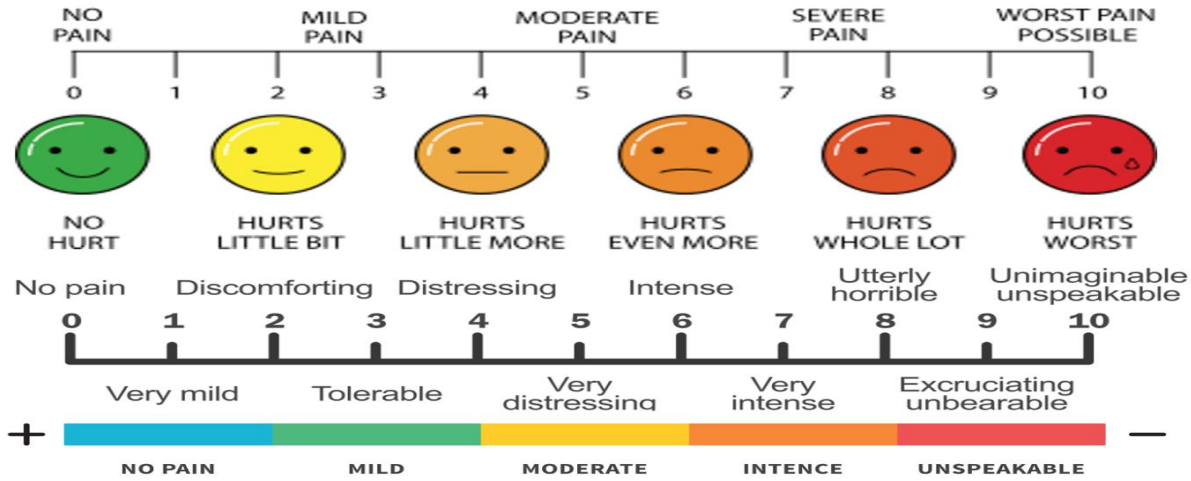
Multi language Wong Baker and Numeric tools are available if needed

https://www.briggshealthcare.com/Wong-Baker-Faces-Pain-Rating-Scale-8-Languages

2. Visual Analogue Scale

How to use?

This tool was created for children aged 8 and older, to Ask the child using numbers from 0 = no pain through to 10 being the worst pain



Physiological indicators

- *Heart rate may increase*
- *Respiratory rate and pattern may shift from normal i.e.: increase, decrease or change pattern*
- *Blood pressure may increase*
- *Oxygen saturation may decrease*

Physiological indicators in isolation cannot be used as a measurement for pain. A tool that incorporates physical, behavioural and self-report is preferred when possible.

Key considerations

- *Assess pain using a developmentally and cognitively appropriate pain tool*
- *Reassess pain after interventions given to reduce pain (e.g., Analgesia) have had time to work*
- *Assess pain at rest and on movement*
- *investigate higher pain scores from expectation*
- *Document pain scores*
- *Use parent/guardian pain behaviour knowledge for children with cognitive impairment.*

OTHER MEASURE OF PAIN

Pupil Size

Pupil assessment cannot be used to assess whether a patient has satisfactory pain relief. However, it has been used in some studies as one measure of the pharmodynamic effects of opioids and blood concentrations of the drug (Fedder et al., 1984, Brokjaer et al., 2015). Therefore, it may be a simple way of measuring the central nervous system (CNS) effects of opioids. If a patient reports inadequate analgesia but has very small pupils (assessed without direct light). It is reasonable to explain to the patient that this means the drug may already be exerting its near-to-maximal effect in the CNS and that different analgesic strategies may be indicated, as further increases in opioid dose may not be safe and/or at least some of the pain may not be responsive to opioids

Patient Satisfaction

Assessment of patient satisfaction are often used as an indicator of “good” and “bad” pain relief. However, they are really more a measure of the patient’s overall satisfaction with their treatment. They can be influenced by factors other than pain intensity, such as expectation of pain, interference with functioning, analgesia-related adverse effects, and the relationship with medical and nursing staff, for example, ability to communicate well, kindness, care shown, and information given (Schug et al, 2020). People may report high levels of satisfaction even though they have moderate-to-severe acute pain.

EFFECT OF PSYCHOLOGICAL FACTORS OF ACUTE PAIN

Pain is an individual and subjective experience to which psychological and social factors contribute. That is, a number of psychological, behavioural, environmental and social factors may influence a patient's response to pain and pain therapy. These factors are important to acute as well as chronic pain settings and in the transition from acute to chronic pain.

Much of the work looking at the effects of various psychological factors on acute pain intensity and opioid use has focused on patients after surgery. In the early post operative period, preoperative anxiety, depression catastrophizing, and emotional distress levels have been shown to correlate with higher pain intensity and or opioid requirements (Jackson et al, 2016; Sobol-Kwapinska et al, 2016; Yang et al, 2019; Horn et al, 2020). The same factors have been cited as predictors of the risk of developing persistent pain after surgery and persistent postoperative opioid use. They have also been associated with widespread reports of pain and persistent of pain after acute musculoskeletal trauma (Macintyre et al, 2014). Anxiety, catastrophising, and depression are potentially modifiable psychological factors. Where possible, a patient may benefit if these factors are identified and addressed before surgery (Levy et al., 2020). At risk patients are identified during their Pre-Operative Assessment Clinic appointment (POAC), the POAC pharmacist highlight the patients to the Acute Pain Service via email.

Anxiety and depression also seem to be important variables affecting Patient Controlled Analgesia (PCA). Patients with anxiety or depression may have higher postoperative pain intensities and may make more PCA demands, including more "unsuccessful" demands made during the lockout interval (Schug et al, 2020).

Untreated pain can also lead to or increase, patient anxiety, fear, sleeplessness, and fatigue. Aggressive behaviour may also be a sign of anxiety and distress

ASSESSMENT OF ADVERSE EFFECTS

To individualise treatments and to maximise patient safety, there needs to be an ongoing assessment of any adverse effects that might be related to pain management therapies. The adverse effects and complications that can result from the treatment of acute pain will vary according to the drug and the technique used. A number of parameters are monitored routinely in the acute pain setting to minimise the risk to patients. These include signs of excessive opioid doses, regardless of route of administration, and early signs of complication related to regional analgesia, particularly epidural analgesia.

OPIOID-INDUCED VENTILATORY IMPAIRMENT

The use of opioids in the treatment of acute pain, can result in significant morbidity and mortality resulting from their adverse effect on ventilation. Opioids cause dose-dependent depression on ventilation, usually referred to as respiratory depression. The term opioid Induced Ventilatory Impairment (OIVI) is a more appropriate term as there are three aspects of ventilatory impairment that must be considered. (Macintyre et al, 2011). These are central depression of the respiratory centre, which can lead to decreased respiratory rate and/or tidal volume, depression of the CNS, leading to sedation, depression of subglottic airway muscle tone, leading to upper airway obstruction. This may lead to increased carbon dioxide levels (Levy, et al, 2020).

Generally, monitoring of patients with OIVI fall into two groups – measures that do not require electronic equipment (respiratory rate, level of sedation, rates of naloxone administration) and those that do (pulse oximetry and carbon dioxide levels). Carbon dioxide levels is the only accurate way to detect OIVI in the clinical setting.

Measurement of Sedation Score and Respiratory Rate

Although commonly used, a decrease in respiratory rate is a late and unreliable sign of OIVI and a normal rate may coexist with significant OIVI (Macintyre et al, 2011). As significant OIVI is always accompanied by sedation, the best indicator is increasing sedation – a general measure of increasing CNS depression. This is monitored used a simple sedation score reflecting a sensible progression of the depression (Appendix 5).

Sedation scores should be monitored regularly in any patient given an opioid for managing their acute pain (Fredrickson & Lambrecht, 2018). A respiratory rate <8 breaths a minute is considered to indicate OIVI.

Oxygen Saturation

Care must be taken in the interpretation of SpO2 readings as OIVI is only one possible cause of low oxygen levels (hypoxemia), particularly in the post-operative setting. Risk factors for postoperative hypoxemia include advanced age, upper abdominal surgery, thoracic operations, major joint replacement, obesity, pre-existing lung disease lung disease, smoking, and severe pain.

While it is recommended that high risk patients are monitored more closely, all patients must be assumed to be at risk of OIVI, and must be assessed at appropriate and appropriate levels (Levy et al, 2020).

REFERENCES

Brokjær, A., Olesen, A. E., Kreilgaard, M., Graversen, C., Gram, M., Christrup, L. L., & Drewes, A. M. (2015). Objective markers of the analgesic response to morphine in experimental pain research. *Journal of Pharmacological and Toxicological Methods*, *73*, 7-14.

Fedder, I. L., Vlasses, P. H., Mojaverian, P., Rocci Jr, M. L., Rotmensch, H. H., Swanson, B. N., & Fergusonx, R. K. (1984). Relationship of morphine-induced miosis to plasma concentration in normal subjects. *Journal of pharmaceutical sciences*, *73*(10), 1496-1497.

Frederickson, T. W., & Lambrecht, J. E. (2018). Using the 2018 guidelines from the joint commission to kickstart your hospital's program to reduce opioid-induced ventilatory impairment. *APSF Newsletter*, *33*(1), 1-8.

Horn, A., Kaneshiro, K., & Tsui, B. C. (2020). Preemptive and preventive pain psychoeducation and its potential application as a multimodal perioperative pain control option: a systematic review. *Anesthesia & Analgesia*, *130*(3), 559-573.

Horn, A., Kaneshiro, K., & Tsui, B. C. (2020). Preemptive and preventive pain psychoeducation and its potential application as a multimodal perioperative pain control option: a systematic review. *Anesthesia & Analgesia*, *130*(3), 559-573.

Joint Commission (2017) R3 report Issue 11: Pain Assessment and Management Standards for Hospitals. Available from: https://www.jointcommission.org/standards/r3-report/r3-report-issue-11-pain-assessment-and-management-standards-for-hospitals/

Levy, N., Quinlan, J., El-Boghdadly, K., Fawcett, W. J., Agarwal, V., Bastable, R. B., ... & Macintyre, P. E. (2021). An international multidisciplinary consensus statement on the prevention of opioid-related harm in adult surgical patients. *Anaesthesia*, *76*(4), 520-536.

Levy, N., Sturgess, J., & Mills, P. (2018). "Pain as the fifth vital sign" and dependence on the "numerical pain scale" is being abandoned in the US: why?. *British journal of anaesthesia*, *120*(3), 435-438.

Macintyre, P. E., Loadsman, J. A., & Scott, D. A. (2011). Opioids, ventilation and acute pain management. *Anaesthesia and intensive care*, *39*(4), 545-558.

Schnabel, A., Yahiaoui-Doktor, M., Meissner, W., Zahn, P. K., & Pogatzki-Zahn, E. M. (2020). Predicting poor postoperative acute pain outcome in adults: an international, multicentre database analysis of risk factors in 50,005 patients. *Pain reports*, *5*(4), e831.

Schug, S. A., Palmer, G. M., Scott, D. A., Halliwell, R., & Trinca, J. (Eds.). (2020). *Acute pain management: scientific evidence* (Vol. 1). Melbourne, Australia: Australian and New Zealand College of Anaesthetists.

Schug, S. A., Palmer, G. M., Scott, D. A., Halliwell, R., & Trinca, J. (Eds.). (2020). *Acute pain management: scientific evidence* (Vol. 1). Melbourne, Australia: Australian and New Zealand College of Anaesthetists.

Sobol-Kwapinska, M., Bąbel, P., Plotek, W., & Stelcer, B. (2016). Psychological correlates of acute postsurgical pain: A systematic review and meta-analysis. *European Journal of Pain*, *20*(10), 1573-1586.

van Boekel, R. L., Vissers, K. C., van der Sande, R., Bronkhorst, E., Lerou, J. G., & Steegers, M. A. (2017). Moving beyond pain scores: Multidimensional pain assessment is essential for adequate pain management after surgery. *PLoS One*, *12*(5), e0177345.

Warden V, Hurley AC, Volicer L. Development and psychometric evaluation of the pain assessment in advanced dementia (PAINAD) scale. J Am Med Dir Assoc. 2003;4:9-15.

Yang, M. M., Hartley, R. L., Leung, A. A., Ronksley, P. E., Jetté, N., Casha, S., & Riva-Cambrin, J. (2019). Preoperative predictors of poor acute postoperative pain control: a systematic review and meta-analysis. *BMJ open*, *9*(4), e025091.

APPENDIX

Appendix 1: Basic Elements of a Pain History

Site of Pain	Location
Circumstances associated with pain onset	• Time of onset • Precipitating events (including details of trauma or surgical procedure
Character of pain	• Sensory descriptors, e.g., sharp, throbbing aching • Neuropathic pain characteristics, e.g., burning, shotting, stabbing
Intensity of pain	• At rest and on movement • Duration • Whether it is continuous or intermittent • Any aggravating or relieving factors
Associated symptoms	• Nausea, sweating
Evaluation of function	• Effect of pain on mobility, activities, and sleep.
Current and prior treatments for pain	• Doses of current and previous medicines including analgesic drugs, frequency of use, efficacy, side effects. • Other non-drug treatments • Health professionals consulted.
Relevant medical history	• Prior or coexisting pain or medical conditions and treatment outcomes
Other patient factors	• Beliefs concerning the cause of pain • Knowledge, expectations, and preference for pain management • Expectations of outcome of pain treatment • Typical coping response for stress or pain, including presence of anxiety disorders (e.g., depression or psychosis) • Family expectations and beliefs about pain, stress, and post operative or post-injury course.

Adapted from (Macintyre & Schug. 2020)

Appendix 2: Functional Activity Scores

Score	Comments
A	No limits of relevant activity
B	Mild limitation of relevant activity
C	Severe limitation of relevant activity

Adapted from Scott & McDonald, 2008

Appendix 3 - The Leeds Assessment of Neuropathic Symptoms and Signs

The Leeds Assessment of Neuropathic Symptoms and Signs (LANSS) pain scale is an assessment tool developed to analyse and classify pain. Its main function is to determine whether nerve damage is the main source of the pain. This is the only tool that is verified to assess the difference between neuropathic and nociceptive pain. It is used to assist clinicians with assessing pain severity or causes.

The LANSS is composed of seven items, which are broken up into two parts. The first portion is a series of five symptom-related questions, while the last part consists of two clinical examination items that may be completed with a health care professional. This pain scale includes the following:

1. Does the pain produce unpleasant sensations, such as tingling, pricking, or pins and needles?
   Yes (5)
   No (0)
2. Is there a different skin aspect in the painful areas, such as redder skin or mottled appearance?
   Yes (5)
   No (0)
3. Does stroking the skin in the painful area or wearing tight clothing items produce unpleasant sensations?
   Yes (3)
   No (0)
4. Are there any sensations, such as electric shocks, bursting, or jumping corresponding to painful episodes (i.e., unexplained bursts of pain)?
   Yes (2)
   No (0)
5. Are there burning sensations in the painful areas or a sudden temperature change?
   Yes (1)
   No (0)
6. Result of stroking the non-painful area and the described painful area with cotton wool.
   Allodynia in painful area (5)
   Normal sensations in both areas (0)
7. Result to touching (pinprick) both areas with a 23-gauge needle.
   Altered PPT in the painful area (3)
   Equal sensations in both areas (0)

A score of 12 or more on this scale confirms a diagnosis of neuropathic pain to some degree. Scores below 12 are less likely to show neuropathic pain

Appendix 4 -Pain Assessment in Advanced Dementia (PAINAD) Scale

Items	1	2	3	Score
Breathing independent of vocalisation	Normal	Occasional laboured breathing. Short periods of ventilation	Noisy laboured breathing. Long periods of hyperventilation Cheyne-Stokes respiration	
Negative vocalisation	None	Occasional moan or groan. Low level speech with a negative disappointing quality	Repeated troubled calling out Load moaning or groaning Crying	
Facial expression	Smiling or inexpressive	Sad, frightened, frown	Facial grimacing	
Body Language	Relaxed	Tense, Distressed, Pacing, Fidgeting	Rigid. Fist clenched Knees pulled up Pulling or pushing away Striking out	
Consolability	No need to console	Distracted or reassured by voice or touch	Unable to console, distract or reassure	
TOTAL				

Warden et al, (2003)

*Five-item observational tool (see the description of each item below).

**Total scores range from 0 to 10 (based on a scale of 0 to 2 for five items), with a higher score indicating more severe pain (0="no pain" to 10="severe pain").

Breathing

1. Normal breathing is characterized by effortless, quiet, rhythmic (smooth) respirations.
2. Occasional laboured breathing is characterized by episodic bursts of harsh, difficult or wearing respirations.
3. Short period of hyperventilation is characterized by intervals of rapid, deep breaths lasting a short period of time.
4. Noisy laboured breathing is characterized by negative sounding respirations on inspiration or expiration. They may be loud, gurgling, or wheezing. They appear strenuous or wearing.

5. Long period of hyperventilation is characterized by an excessive rate and depth of respirations lasting a considerable time.
6. Cheyne-Stokes respirations are characterized by rhythmic waxing and waning of breathing from very deep to shallow respirations with periods of apnoea (cessation of breathing).

Negative vocalization

1. None is characterized by speech or vocalization that has a neutral or pleasant quality.
2. Occasional moan or groan is characterized by mournful or murmuring sounds, wails or laments. Groaning is characterized by louder than usual inarticulate involuntary sounds, often abruptly beginning and ending.
3. Low level speech with a negative or disapproving quality is characterized by muttering, mumbling, whining, grumbling, or swearing in a low volume with a complaining, sarcastic or caustic tone.
4. Repeated troubled calling out is characterized by phrases or words being used over and over in a tone that suggests anxiety, uneasiness, or distress.
5. Loud moaning or groaning is characterized by mournful or murmuring sounds, wails or laments much louder than usual volume. Loud groaning is characterized by louder than usual inarticulate involuntary sounds, often abruptly beginning and ending.
6. Crying is characterized by an utterance of emotion accompanied by tears. There may be sobbing or quiet weeping.

Facial expression

1. Smiling is characterized by upturned corners of the mouth, brightening of the eyes and a look of pleasure or contentment. Inexpressive refers to a neutral, at ease, relaxed, or blank look.
2. Sad is characterized by an unhappy, lonesome, sorrowful, or dejected look. There may be tears in the eyes.
3. Frightened is characterized by a look of fear, alarm or heightened anxiety. Eyes appear wide open.
4. Frown is characterized by a downward turn of the corners of the mouth. Increased facial wrinkling in the forehead and around the mouth may appear.
5. Facial grimacing is characterized by a distorted, distressed look. The brow is more wrinkled as is the area around the mouth. Eyes may be squeezed shut.

Body language

1. Relaxed is characterized by a calm, restful, mellow appearance. The person seems to be taking it easy.
2. Tense is characterized by a strained, apprehensive or worried appearance. The jaw may be clenched (exclude any contractures).
3. Distressed pacing is characterized by activity that seems unsettled. There may be a fearful, worried, or disturbed element present. The rate may be faster or slower.
4. Fidgeting is characterized by restless movement. Squirming about or wiggling in the chair may occur. The person might be hitching a chair across the room. Repetitive touching, tugging or rubbing body parts can also be observed.
5. Rigid is characterized by stiffening of the body. The arms and/or legs are tight and inflexible. The trunk may appear straight and unyielding (exclude any contractures).
6. Fists clenched is characterized by tightly closed hands. They may be opened and closed repeatedly or held tightly shut.
7. Knees pulled up is characterized by flexing the legs and drawing the knees up toward the chest. An overall troubled appearance (exclude any contractures).
8. Pulling or pushing away is characterized by resistiveness upon approach or to care. The person is trying to escape by yanking or wrenching him or herself free or shoving you away.
9. Striking out is characterized by hitting, kicking, grabbing, punching, biting, or other form of personal assault.

Consolability

1. No need to console is characterized by a sense of well-being. The person appears content.
2. Distracted or reassured by voice or touch is characterized by a disruption in the behaviour when the person is spoken to or touched. The behaviour stops during the period of interaction with no indication that the person is at all distressed.
3. Unable to console, distract or reassure is characterized by the inability to sooth the person or stop a behaviour with words or actions. No amount of comforting, verbal or physical, will alleviate the behaviour.

Appendix 5 – Assessment of Opioid-Induced Ventilatory Impairment (OIVI)

Sedation scores	0 = wide awake 1 = easy to rouse 2= easy to rouse but unable to stay awake; early OIVI 3 = somnolent, difficult to rouse, severe OIVI.
Respiratory rate	Less than 8 breath/min is often considered to be a sign of OIVI, but generally an unreliable indicator OIVI can coexist with a normal respiratory rate.
Oxygen saturation	May also be unreliable, especially if the patient is receiving supplemental oxygen. Oxygen saturations may be low for reasons other than OIVO Unless Pulse oximetry monitoring is continuous, intermittent episodes of low oxygen saturations may be missed
End-tidal or transcutaneous carbon dioxide levels	The most sensitive and accurate way of detecting OIVI in the clinical setting. Only used in critical care and recovery,